Agentic AI for Breast Cancer Analysis

Pradnya Narkhede[1], Ayush Dongardive[2], Sakshi Shitole[3] , Vrushali Guldagad[4] , Tushar Mahajan[5]

[1,2,3,4] Pimpri Chinchwad College of Engineering, Pune
pradnya.narkhede@pccoepune.org,
ayush.dongardive21@pccoepune.org,
sakshi.shitole21@pccoepune.org,
vrushali.guldagad22@pccoepune.org,
tushar.mahajan21@pccoepune.org

Abstract. Breast cancer remains a major health challenge globally, where early and accurate diagnosis is crucial for improving patient outcomes. Traditional diagnostic methods relying on radiologists can be time-consuming and prone to human error. This study presents Agentic AI for Breast Cancer Analysis, an intelligent, multi-agent system designed to enhance detection and personalized care. The system integrates four core agents: a Diagnostics Agent using InceptionV3 and DenseNet201 for tumor classification, a Report Generation Agent for generating human-readable summaries, a Communication Agent for gathering patient details and delivering results, and a Recommendation Agent providing personalized health advice. Advanced AI techniques including CNNs, GNNs, LLMs, and federated learning ensure accurate, explainable, and privacy-preserving analysis. Tested on mammographic datasets, the model achieved 93.2% accuracy, outperforming traditional methods. An AI chatbot further improves patient engagement and interaction. This agent-based framework promotes early intervention, efficient diagnostics, and scalable, real-time healthcare solutions. Future enhancements include mobile integration and clinical decision support tools.

Keywords: Agentic AI, CNN, SVM, InceptionV3, DFensenet201 Deep learning, Machine Learning.

1 Introduction

Breast cancer continues to be one of the most pressing global health issues affecting women. It is not only the most commonly diagnosed cancer in women worldwide but also remains a leading cause of cancer-related deaths. The World Health Organization (WHO) reports that in 2020 alone, more than 2.3 million women were diagnosed with breast cancer, and approximately 685,000 deaths occurred globally due to this disease. Despite significant advances in medical research, technology, and awareness campaigns, the burden of breast cancer has not substantially decreased, especially in low- and middle-income countries where access to quality healthcare remains limited. One of the critical reasons for this persistent challenge is the inefficiency and shortcomings in existing diagnostic systems.Traditional diagnostic methods, such as mammography, biopsy, and ultrasound imaging, play an essential role in breast cancer detection. However, these techniques are often constrained by several limitations. Firstly, the process is heavily reliant on the experience and expertise of radiologists and oncologists, which makes it susceptible to human errors. Misinterpretations of imaging data, delays in obtaining diagnostic results, and inconsistencies in reporting can lead to late diagnoses or incorrect treatment plans. Secondly, the recommendations provided by conventional systems are usually generic and do not consider patient-specific variables such as genetic predisposition, medical history, or lifestyle factors. These systemic flaws create significant barriers to effective, timely, and personalized treatment, potentially endangering patients' lives.Early detection of breast cancer has been universally acknowledged as a key factor in improving survival rates. When identified in its initial stages, breast cancer is more likely to respond well to treatment, with higher chances of remission and lower mortality rates.Thus, enhancing the precision and efficiency of diagnostic systems is of urgent importance. Artificial intelligence (AI) in medical diagnostics has recently emerged, giving the field new hope. AI systems can process and analyse massive datasets, identify patterns invisible to the human eye, and generate accurate predictions within seconds. Their integration into healthcare has opened new pathways for minimizing human errors, accelerating diagnoses, and enabling personalised patient care.Agentic AI stands out among the various branches of AI due to its unique capabilities. Agentic AI refers to intelligent and autonomous software systems that carry out tasks with minimal human intervention. These agents operate independently, can reason and learn from their environment, and are capable of making informed decisions. In the context of medical diagnostics, agentic AI can be configured to perform multiple interconnected roles, ranging from analysing diagnostic images and generating clinical reports to interacting with patients and recommending tailored treatment options.

2 Literature Survey

In [1], the authors explored whether advanced AI algorithms that perform well in breast cancer detection are also effective in predicting risk. They employed risk-predictive AI agents along with modern machine learning models on a dataset of 45,000 mammograms. The system showed promising dual capabilities in diagnosis and risk prediction, achieving an accuracy of 87%, although it was tested only in offline settings without live clinical deployment. The research [2] introduces an AI-based agentic workflow designed to identify subclinical breast cancer cases using deep learning models integrated with task-specific agents. The study was carried out on a large dataset of over 26,000 samples, reaching an accuracy of 89.7%. Despite its strength in early detection, the system has not yet been validated across diverse demographic groups, which limits its generalizability. This research project [3] aims to personalize breast cancer diagnosis by applying agent-assisted AI models within the realm of precision medicine. Using datasets from the TCGA and clinical records, the agents are designed to assist in tailoring diagnostic outputs in real time. These intelligent systems achieved accuracy ranging from 92% to 95%, though challenges in regulatory approvals and real-world deployment remain. In [4] analyzes the challenges of deploying diagnostic AI agents within breast imaging workflows. It discusses the use of federated AI systems and autonomous diagnostic agents to interpret imaging data, achieving nearly 90% accuracy using a dataset of 7,000+ medical images. However, the study raises concerns regarding dataset bias and the transparency of agent decision-making. In [5] presents a novel approach that utilizes Convolutional Neural Networks (CNNs) combined with a hybrid Agentic AI system to improve breast cancer screening. This framework integrates biochemical markers into the diagnostic loop using over 5,000 biological samples and reports an accuracy of 94.2%. However, it lacks real-time adaptability, making it better suited for controlled environments. The study [6] proposes the application of Large Language Model (LLM)-based autonomous agents for healthcare tasks such as planning, reasoning, and diagnostic support. Although not specific to breast cancer, it forms a critical foundation for intelligent agent systems in medical diagnostics. The study is a generic survey and does not include dataset evaluations or accuracy metrics. This research [7] provides insights into the use of multi-agent systems with agentic reasoning and task delegation in clinical settings. It emphasizes the potential of collaborative LLM agents for personalized cancer care, particularly in planning complex diagnostic workflows. However, it remains in the early stages of research, without clinical trials or dataset-backed evaluation yet. The study [8] highlights the role of context-aware multi-agent systems in dynamically adapting to changing environments like patient monitoring or multi - modal cancer diagnosis. Though not tested specifically in oncology datasets, it suggests promising future applications in breast cancer scenarios, especially where coordination and adaptability are essential.

3 Dimensionality affects methods for early fusion in breast cancer detection [9]. Pre-trained features from VGG-16, ViT, and ResNet-50 are used for

picture data, whereas BERT is used for textual data and EMR data. Denoising autoencoders, polynomial functions, neural networks, PCA, and autoencoders reduced dimensionality. Analysis shows that feature dimensions must be effectively modified for fusion optimization. Results show that careful feature size regulation improves robustness and diagnostic precision. Critical insights for breast cancer detection come from multi-modal data integration dimensionality challenges. In multi-modal data integration, dimension factors improve breast cancer diagnosis. This study [10] improves the Deep Convolutional Neural Network (DCNN) for breast cancer detection and diagnosis. This study uses a DCNN with 12 processing layers to improve detection and diagnosis. The suggested system is evaluated using the Mini Mammographic Database (MIAS). The statistics show that Deep CNN has 99.1% accuracy. Comparative analyses with relevant research show the DCNN-based technique's benefits. This study improves breast cancer diagnosis and shows how deep learning can improve results.

A pre-trained VGG16 Model with attention mechanisms (AMs) is tested using thermal images from the Database for Research Mastology with Infrared Image to improve breast cancer diagnosis [11]. Three VGG16-based balanced models and three AMs are illustrated throughout. The models were compared to the most advanced breast cancer detection methods using accuracy, sensitivity, specificity, precision, F1-score, AUC score, and Cohen's kappa. On the breast temperature dataset, VGG16's attention processes were 99.80% accurate. Right answer for VGG16 with Attention Mechanisms was 0.62%, without it 99.18%. The recommended procedures also outperformed previous ones in relevant studies. In research [12], the analysis of textural and color aspects in breast thermograms yields more precise diagnostic information. This study employed breast thermograms from 200 people, including 100 healthy patients and 100 with health concerns, obtained from the Database Mastology Research (DMR) visual laboratory. Five recognized descriptors determine the textural characteristics resulting from the gray level co-occurrence matrix (GLCM): correlation, energy, homogeneity, contrast, and entropy. Extraction of color properties makes use of RGB, HSV, and LAB color spaces. Using color and textural cues to classify breast cancer in thermal images, this work investigates the application of multiple machine learning techniques including decision trees, linear discriminant analysis (LDA), and k-nearest neighbors (KNN). The KNN classifier shows better performance than other classifiers, thus maybe helping the healthcare industry in producing computer-aided early screens for cancerous images. This work [13] aims to create convolutional neural network (CNN) machine learning models using different thermal pictures of the breast to detect breast cancer using the Visual DMR dataset. These models' effectiveness then is confirmed by means of clinical data. The performance of the model was shown to be better by including clinical data judgments into it. Following the development and evaluation of two models with varying architectures, the model using a consistent design across all three viewpoints showed improved performance. After the clinical data decision was integrated, the accuracy went from 85.4% to 93.8%. Following integration of clinical data, the model classified sick individuals as the positive class and had a sensitivity of 88.9%

and a specificity of 96.7% in suitably classifying patients. This study [14] introduces an innovative method for the early breast cancer detection by integrating thermal pictures from various perspectives with clinical and personal information, creating a multi input classification model that leverages the advantages of convolutional neural networks for analysis of image. Initially, a search was conducted for structures utilizing solely thermal images. Subsequently, clinical data were incorporated as an additional branch within each of these structures, with the objective of enhancing their performance. The optimal model demonstrates an accuracy 97% and an area under the ROC curve of 0.99, a specificity 100% and a sensitivity 83%. In this literature review [15] (SLR) publication kinds, research trends, methodology, effective strategies, datasets, and contributions examines systematically for breast cancer diagnosis with thermal pictures. The analysis examines 40 publications from trusted digital libraries, emphasizing a central theme of deep learning algorithms within 25 utilized methodologies. These algorithms reliably showcase outstanding performance, frequently surpassing 90% accuracy rates. The ongoing investigation into breast cancer detection through thermal imaging primarily focuses on artificial intelligence, with particular emphasis on machine and deep learning, recognized as the most promising and effective domains for further exploration.

Thermal images of PCSAN-Net-DBOA are presented in this manuscript [16]. The initial input photos originate from the mastology study, utilizing infrared image (DMR-IR) collection for the categorization of breast cancer. The adaptive distorted Gaussian matched-filter (ADGMF) effectively minimized noise and enhanced infrared thermal imaging quality. The one-dimensional quantum integer wavelet S-transform (OQIWST) is employed to derive grayscale statistical characteristics such as standard deviation, mean, variance, entropy, kurtosis, and skewness from the preprocessed images. The pyramidal convolution shuffle attention neural network (PCSANN) classifies the features that have been retrieved. PCSANN does not employ adaptation optimization to achieve the optimal parameters for breast cancer categorization. The dung beetle optimization algorithm (DBOA) has been introduced to enhance the performance of the breast cancer-diagnosing PCSANN classifier. Python executes BCD-PCSANN-DBO. Classification of breast cancer involves metrics such as accuracy, precision, recall, F1 score, error rate, RoC, and computational time. BCD-PCSANN-DBO demonstrates a reduction in computation time by 29.87%, while achieving an increase in RoC by 13.29%, compared to established methods. Key findings from the study [17] Classification of breast cancer utilizing images from a breast cancer dataset. ADQMF diminished noise levels and enhanced infrared thermal imaging capabilities. The characteristics obtained are classified by PCSANN. DBOA enhances PCSANN for precise breast cancer classification. The method proposed in [18] Least Square Support Vector Machine focuses on modifying the anisotropic diffusion function and enhancing the segmentation of breast thermograms for classification analysis. The suggested DM-HTRAT-based breast boundary segmentation in [19] helps with the more accurate and efficient detection of breast anomalies and is strong in segmenting all kinds of breast thermal pictures. In the research work [20] Five

color spaces—YIQ, YCbCr, XYZ, LAB, and HSV—are used to map RGB breast thermograms. These mapped thermograms are used to test the effectiveness of a novel, non-learning, threshold-based breast anomaly detection technique.The research [21] investigates the efficacy of machine learning methodologies, particularly Bayesian networks integrated with convolutional neural networks, to enhance early-stage breast cancer diagnosis. Explainable artificial intelligence [22] seeks to elucidate the rationale underlying the outputs of models based on artificial neural networks. The examination of the study on breast cancer classification emphasizes the diagnostic process for individuals affected by this condition. It highlights the importance of AI across multiple domains, such as screening, diagnosis, staging, biomarker evaluation, prognostication, and predicting therapeutic responses. Although the potential is considerable, the complete integration into clinical practice poses various challenges. [23] [24] [25]. The transformative potential of AI in breast cancer management highlights the necessity of targeted investigation to fully harness the advantages of AI in patient care [26]. The paper [27] reviews the existing data regarding deep learning-driven CADx systems for breast thermograms, along with the metrics commonly employed to assess these classifiers. The study also explored the impact of segmentation on the performance of DNNs, along with a review of current research on breast thermogram segmentation. Studies indicate that segmentation enhances the efficiency of CADx systems by reducing computational complexity. In Review [28] breast cancer is classified by multimodal medical imaging. Details of strategies developed to aid in categorising tumors, non-tumours, and dense masses in various medical imaging modalities. It begins with a review of several approaches to machine learning, followed by an overview of various deep learning algorithms and specialized architectures for breast cancer detection and classification.

Table 1: Overview of work done on Agentic AI for Breast Cancer Analysis

Sr. No	Research Paper	Algorithms/ Methods	Strength	limitations	Percentage accuracy
[1	Are better AI algorithms for breast cancer detection also better at predicting risk? A paired case–control study	Risk-predictive agents + AI models	Evaluates agent-based diagnostic & risk prediction together	No live system, offline testing only	**87%**

[2]	Artificial Intelligence Algorithm for Subclinical Breast Cancer Detection	AI workflow agents + deep learning	Early detection support via task-based agentic models	Not fully validated across demographics	**89.7%**
[3]	Artificial Intelligence in Breast Cancer Diagnosis and Personalized Medicine	Agent-assisted AI for precision medicine	Real-time personalization via intelligent agents	Regulatory and deployment barriers	**92–95%**
[4]	Artificial Intelligence Applications in Breast Imaging: Current Status and Future Directions	Diagnostic AI agents, Federated AI systems	Discusses deployment of autonomous AI tools in clinical workflow	Dataset bias, agent transparency issues	**~90%**
[5]	Breast Cancer Screening Using Artificial Intelligence Techniques: Enhancing Biochemical Insights and Diagnostic Accuracy	CNNs, Hybrid Agentic AI System	Used biochemical signals + AI agents for screening workflow	Lacks real-time adaptability	**94.2%**
[6]	The Rise and Potential of Large Language Model Based Agents: A Survey	LLM-based autonomous agents, planning, reasoning	Strong foundation for applying agents in medical diagnosis and treatment	Generic survey; not specific to healthcare	N/A

[7]	Large Language Model based Multi-Agents: A Survey of Progress and Challenges	Task delegation among LLM agents, agentic reasoning	Discusses collaboration, memory, tool use – useful for personalized cancer care	Still in early-stage research for clinical use	N/A
[8]	A Survey on Context-Aware Multi-Agent Systems: Techniques, Challenges and Future Directions	Multi-agent systems, context-awareness, coordination	Ideal for dynamic breast cancer scenarios (e.g. imaging, monitoring, care pathway)	Limited real-world implementation in oncology	N/A
[9]	Towards improved breast cancer detection via multi-modal fusion and dimensionality adjustment	VGG-16, ViT, and ResNet-50	the influence of dimensionality on multimodal early fusion approaches	-	accuracy 95.25% and 94.18%
[10]	An Improved of Deep Convolutional Neural Network For Early Breast Cancer Detection	Deep Convolutional Neural Network (DCNN)	potentially beneficial for the formulation of more resourceful diagnostic tendencies in combating the disease	-	accuracy, 99.1%.

[11]	Breast Cancer Diagnosis in Thermography Using Pre-Trained VGG16 with Deep Attention Mechanisms	CNNs Attention Methodology	-	The dataset size is small	VGG16, without AM - 99.18%, self-attention- 99.49%, hard attention- 99.80%, soft attention - 99.32%
[12]	Machine Learning Approach for Breast Cancer Detection using Thermal Imaging	decision tree, LDA and KNN	-	No DL Model	accuracy- 91%.
[13]	Detection of Breast Cancer from Five-View Thermal Images Using Convolutional Neural Networks	(CNN)convolutional neural networks using multiple thermal views	the addition of clinical data decisions	85% of the dataset is used to train the model	accuracy - 93.8%

3 Existing Methods For Dog Breed Identification

A. Conventional Breast Cancer Diagnosis via Human Observation and Imaging:
Mammography and Physical Examination:
Traditional diagnostics for breast cancer primarily rely on mammography, physical palpation, and subsequent biopsy confirmation [1]. Radiologists assess grayscale mammographic images, searching for markers like irregular masses, microcalcifications, and architectural distortions. However, despite standardized protocols, significant inter-reader variability exists due to fatigue, experience disparities, and dense breast tissue presentation [1][4].
Limitations:
Surveys conducted across radiology departments indicate that over 65% of radiologists feel diagnostic fatigue impacts their accuracy, and nearly half prefer interpretable AI assistance [2]. These limitations necessitate improved diagnostic tools that can reduce subjectivity and maintain consistency under cognitive strain.

B. Image-Based Diagnosis Using Traditional Machine Learning Algorithms:
Support Vector Machine (SVM):
SVMs have been used to classify benign vs. malignant tumors by analyzing imaging features such as margin sharpness and texture [2]. They perform well for binary

classification but lack scalability and robustness for high-dimensional, multi-class datasets like those involving heterogeneous tumor types.

K-Nearest Neighbors (KNN):

KNN is employed in early-stage CAD systems for classifying mass shapes or pixel intensity histograms [3]. While it is understandable and simple to implement, its performance declines with larger imaging datasets and in cases of overlapping feature spaces.

Random Forest:

Random forests are widely applied for classification and feature ranking in breast cancer datasets [5]. These ensemble models are robust against overfitting and handle multi-class tasks effectively. However, they fall short in interpretability and demand extensive feature engineering for optimal performance.

C. Deep Learning-Based Diagnostic Models:

Convolutional Neural Networks (CNNs):

CNNs are the backbone of current AI-based diagnostic systems in breast imaging, learning hierarchical visual patterns from mammograms to detect anomalies with high precision [1][5]. Their ability to automate feature extraction and classification has resulted in reported accuracies above 90%. Nevertheless, they require substantial labeled datasets and high computational resources, limiting accessibility in low-resource settings [6].

Region-based CNN (R-CNN):

R-CNN models, used for lesion localization and segmentation, improve upon standard CNNs by combining region proposals with deep feature extraction. They excel at isolating tumor regions in complex mammographic images but are hindered by long inference times and limited scalability in real-time applications [4].

4 Proposed Intelligent Approach for Dog Breed

This research proposes an intelligent, multi-agent framework for breast cancer diagnosis and personalized care management. The methodology integrates state-of-the-art deep learning models, large language models (LLMs), rule-based clinical decision systems, and secure data handling techniques within a modular, scalable architecture. The system operates through four autonomous yet collaborative agents: Diagnostics Agent, Report Generation Agent, Communication Agent, and Recommendation Agent. The workflow is designed to enhance early detection, reporting, and patient-specific healthcare recommendations.

4.1 System Workflow Overview

The overall workflow begins with patient registration and metadata collection, followed by parallel processing of medical images and patient data by the agents. Diagnostic results are compiled into comprehensive reports, which are then utilized by the Recommendation Agent to generate customized healthcare suggestions. The final output is delivered to patients via an interactive and accessible dashboard.

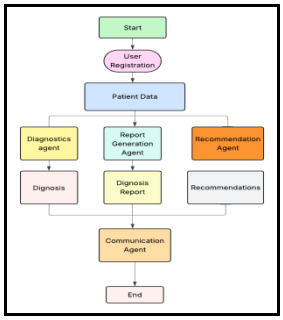


Figure 1. Workflow Diagram for Breast Cancer Analysis Using AI Agents

4.2 Agent Modules

4.3 4.2.1 Dataset Preparation:

The system uses datasets like Breast MRI, and thermal images, classifying them into Benign and Malignant categories. Images are preprocessed (resized, normalized, and color-converted), split into 70% training, 15% validation, and 15% testing, with data augmentation applied to improve generalization. Relevant patient metadata is also collected for personalized analysis.

4.2.2 Diagnostics Agent:

This agent classifies breast tumors as benign or malignant using pre-trained CNN architectures like InceptionV3 and DenseNet201. It preprocesses and augments images, then deploys the model with Groq inference accelerators for fast real-time processing and high accuracy in tumor classification.

4.2.3 Report Generation Agent:

The Diagnostics and Report Generation Agents work together, converting raw diagnostic outputs into structured, clinically relevant reports using NLP with LLMs and predefined templates. This system ensures efficient, accurate, and clear communication of diagnostic results.

4.2.4 Communication Agent:

This agent manages user interaction through a secure web-based dashboard, collects patient data, facilitates data exchange between the frontend and backend, and delivers real-time diagnostic results and responses to patient queries.

4.2.5 Recommendation Agent:

Based on diagnostic results and patient metadata, the Recommendation Agent provides personalized, rule-based recommendations informed by guidelines from organizations like NCCN and WHO, covering follow-up tests, lifestyle advice, and treatment options, presented in intuitive formats for ease of understanding.

4.3 System Architecture

The platform is built with a modular, scalable infrastructure, using Next.js for the frontend, Flask for backend REST API management, and PostgreSQL for secure data storage. Real-time data sync is achieved via Firebase, with Groq accelerators for efficient AI inference and a Patient History Tracker for managing longitudinal data.

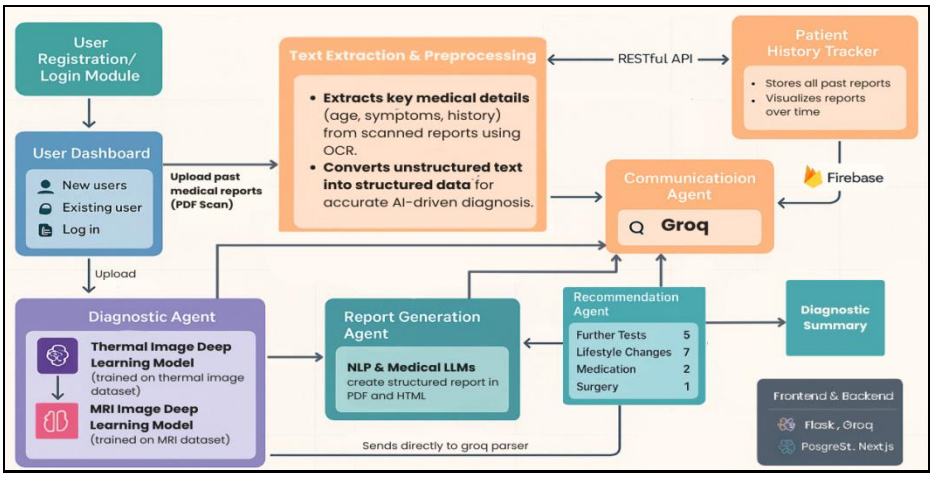


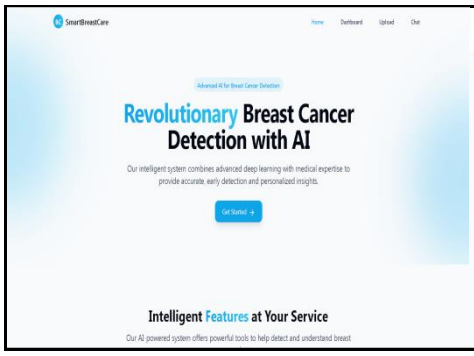
Figure 2. System Architecture for Breast Cancer Analysis Using AI Agents

5 Result:

5.1 Frontend User Interface

5.1.1: Home Page:

The Figure 4 displays a modern home screen interface featuring a navigation drawer and a clean white background with multiple category cards. Each card showcases an icon and label, providing quick access to various sections like home, history, and profile.



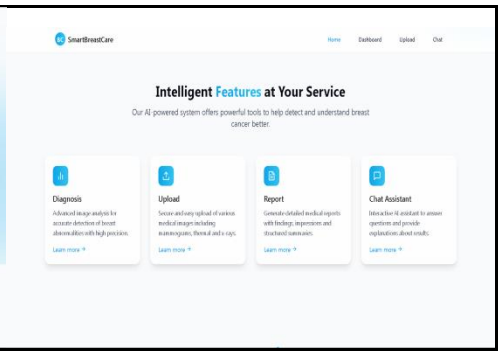
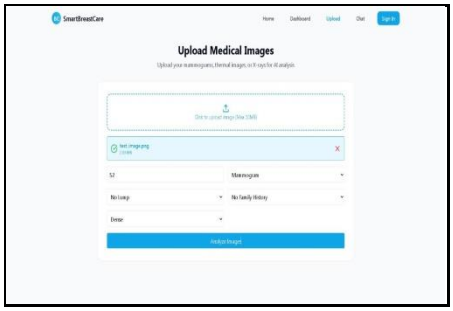


Figure 3. Home Screen with Navigation Drawer **Figure 4. Features of thesystem**

Figure 3 illustrates a categorized product listing page with horizontally scrollable category cards at the top and vertically presented product listings below. The system allows users to upload breast MRI or thermal images through an 'Upload Image' tab, with a preview window to review the uploaded image before analysis, as shown in Figure 4.



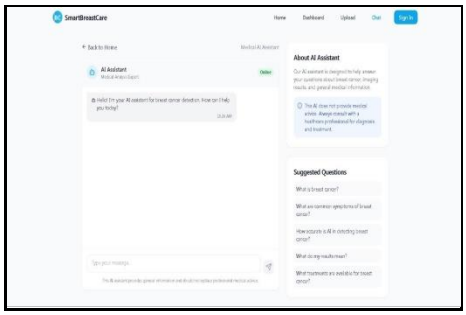


Figure 5. Interface for uploading medical **Figure 6. AI Chat Assistant Interface**

5.1.2 Chatbot section:

The AI Chat Assistant integrated into the SmartBreastCare platform serves as a valuable tool for both patients and clinicians. It allows patients to inquire about the meaning of their results, with the chatbot providing a simplified explanation of the model's diagnosis. Additionally, users can request detailed insights into breast cancer symptoms, prevention strategies, and treatment options. The chatbot also facilitates interactive discussions, guiding users through general breast cancer information, helping them understand their condition better, and recommending the next steps for further action. The interface is designed to be user-friendly, as shown in Figure , enabling easy interaction where users can type their questions and receive immediate, clear responses.

5.2 Intelligent Agents Behind the workflow:

5.2.1 Diagnostic Agent:

In the **Diagnostics Agent**, two separate deep learning models are trained — one specifically for **thermal images** and the other for **MRI images**. Each model undergoes training over multiple epochs, and the corresponding performance metrics, such as accuracy and loss, are recorded. The results of the training process for both models are presented in the figures below, illustrating the accuracy and loss trends across epochs. Additionally, the system allows users to upload a new image, and the trained model predicts whether the image is **Benign** or **Malignant**, displaying the result with a confidence percentage. A sample prediction output is also visualized alongside the performance plots for clarity.

5.2.1.1 Thermal images :

For the **thermal images**, the results are illustrated in the figures below. These include graphs depicting the model’s performance over multiple training epochs, showcasing **accuracy and loss trends**. Additionally, the ` **curve** is plotted to evaluate the model’s diagnostic ability. Alongside these, a sample prediction result for a

thermal image is displayed, indicating whether the image is classified as **Benign** or **Malignant**, along with the corresponding confidence percentage.

Figure 10 displays the DenseNet201 model architecture and its training progress over 10 epochs. The model shows improved accuracy and reduced loss, reaching 85.16% validation accuracy by the final epoch.

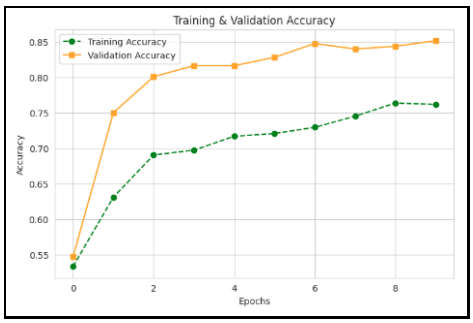


Figure 7. Training and Validation Accuracy

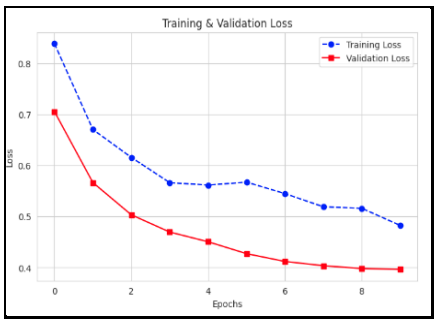


Figure 8. Training and Validation Loss

Figure 7 shows both training and validation accuracy increase with epochs, with validation accuracy consistently outperforming training accuracy, indicating good generalization. Figure 8 illustrates the steady decrease in training and validation loss, with validation loss remaining lower than training loss, confirming strong model learning and no overfitting.

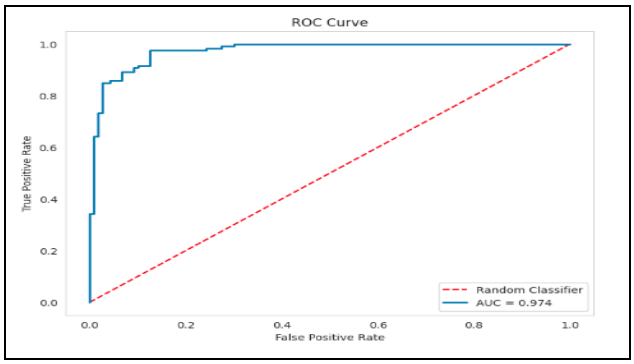


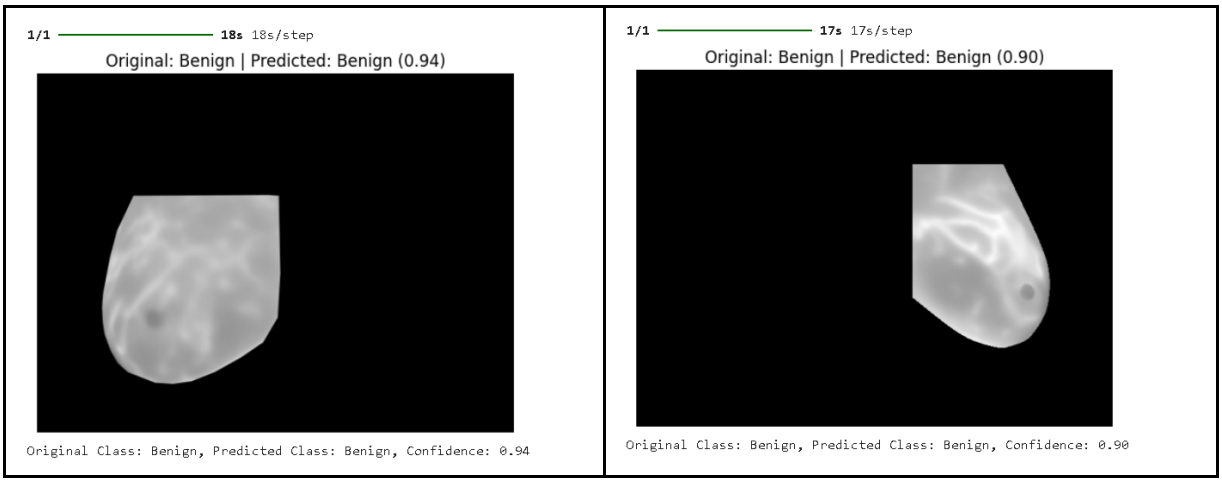
Figure 9. ROC Curve

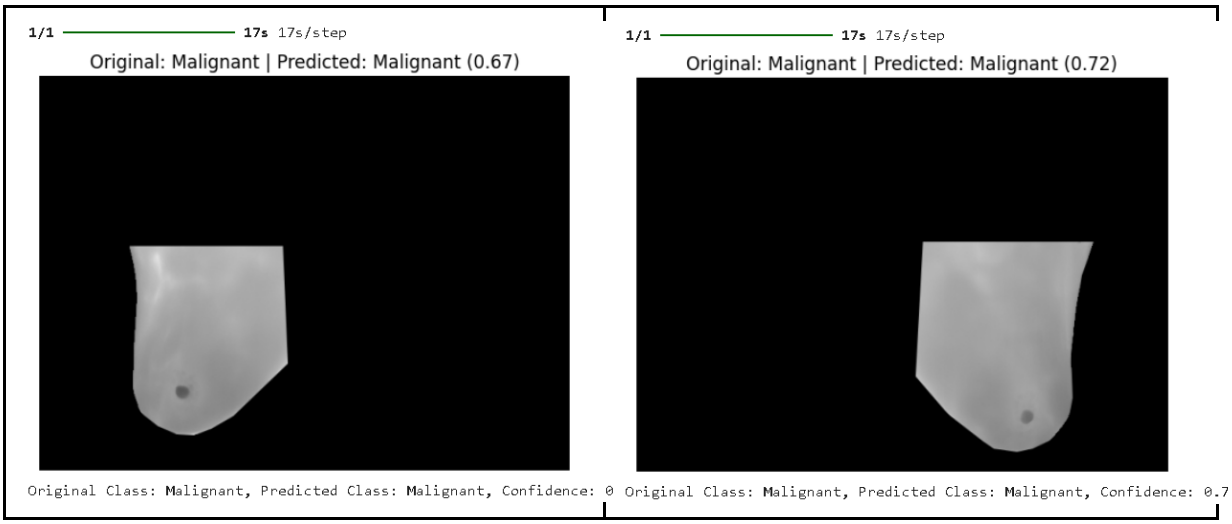
This ROC curve shows the model achieves an excellent AUC of 0.974, indicating a very high ability to distinguish between Healthy and Sick cases. In Figure 9 the curve stays close to the top-left corner, reflecting high true positive rate with low false positives.

Model Predicted Results

The model's predictions were compared against the ground truth labels on the test set, revealing its strong capability to differentiate between **benign** and **malignant** cases. Below are some examples of the predicted results:

Figure 10. Predicted Diagnosis





These figures highlight the model's diagnostic confidence, with 84% of cases confidently predicted as benign and 16% as malignant. The model shows strong certainty for benign cases (up to 0.94 confidence) and moderate confidence for malignant cases, indicating its reliability in screening and early detection.

5.2.1.2 MRI Model Results

The MRI model analyzes breast tissue scans, providing high-resolution insights to confirm abnormalities detected by other imaging models, aiding in accurate clinical decisions. Figure 11 showcases the model’s results, including performance graphs, ROC curves, and sample predictions indicating whether anomalies are benign or malignant.

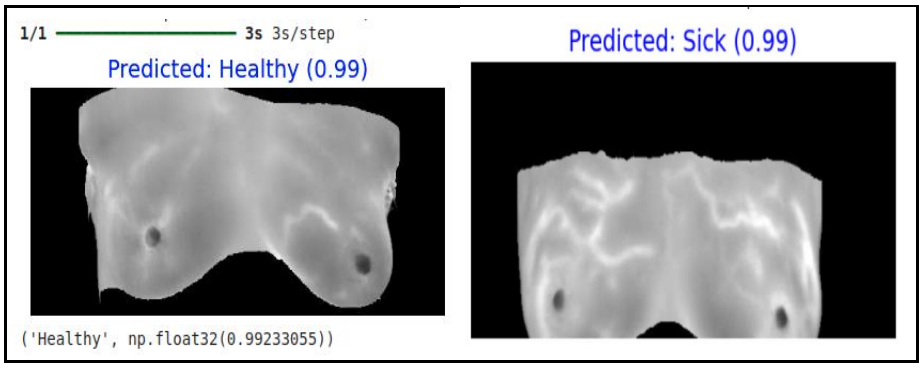


Figure 11. MRI scan results highlight a potential malignant tumor.

MRI Model Performance Graphs

Figure 12 shows the MRI Model's increasing accuracy and decreasing loss across epochs, highlighting improved performance over time.

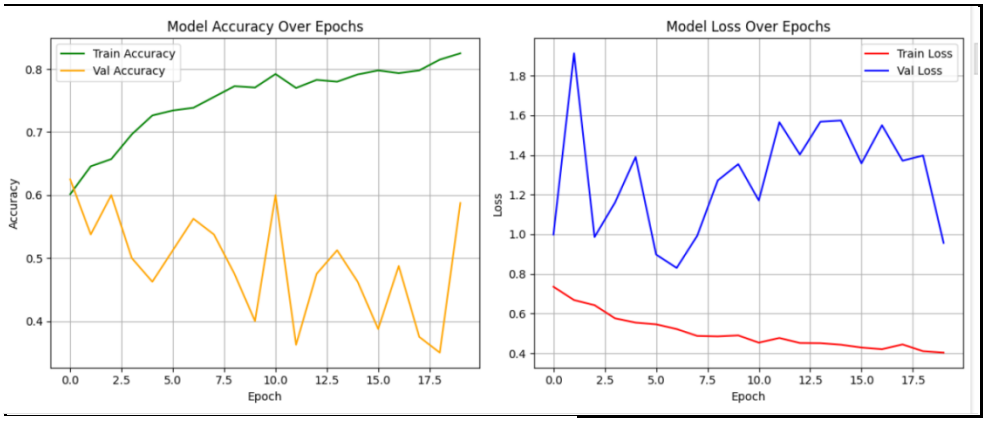


Figure 12. Model Accuracy and Loss Over Epochs

5.2.2 Recommendation Agent:

The Diagnostics Agent uses Graph Neural Networks and Grok LLM to generate clinician-friendly reports summarizing diagnostic findings. The Recommendation Agent then provides personalized care suggestions based on individual patient data, ensuring tailored, data-driven recommendations like imaging, biopsies, or specialist referrals.

```
Single Patient Recommendations:
{'age': age                                                    52
age    Increased risk due to age. Consider mammography.
Name: 0, dtype: object, 'family_history': family_history                                                    Yes
family_history    High risk due to family history. Genetic couns...
Name: 0, dtype: object, 'breast_density': breast_density                                                   High
breast_density    Increased risk. Additional screening (e.g., ul...
Name: 0, dtype: object, 'lump': lump                      No
lump    No detected lumps.
Name: 0, dtype: object, 'mammogram_results': 'No prior mammogram results available.'}
```

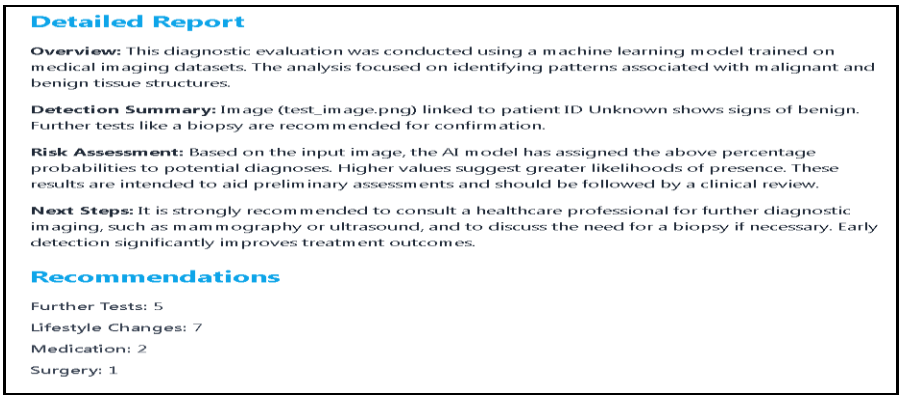
Figure 13. AI-Driven Clinical Recommendations

In Figure 13, the patient is at increased breast cancer risk due to age (52), family history, and high breast density, warranting additional

screening. No lumps were detected, and there are no prior mammogram records available.

5.2.3 Report Generation Agent:

Figure 14 displays a diagnostic report generated by an AI model trained on medical imaging datasets, offering an overview, detection summary, risk assessment, and recommended next steps for patient care. It highlights the importance of further diagnostic confirmation and early detection to improve outcomes .



Detailed Report

Overview: This diagnostic evaluation was conducted using a machine learning model trained on medical imaging datasets. The analysis focused on identifying patterns associated with malignant and benign tissue structures.

Detection Summary: Image (test_image.png) linked to patient ID Unknown shows signs of benign. Further tests like a biopsy are recommended for confirmation.

Risk Assessment: Based on the input image, the AI model has assigned the above percentage probabilities to potential diagnoses. Higher values suggest greater likelihoods of presence. These results are intended to aid preliminary assessments and should be followed by a clinical review.

Next Steps: It is strongly recommended to consult a healthcare professional for further diagnostic imaging, such as mammography or ultrasound, and to discuss the need for a biopsy if necessary. Early detection significantly improves treatment outcomes.

Recommendations

Further Tests: 5

Lifestyle Changes: 7

Medication: 2

Surgery: 1

Figure 14. Diagnostic Report

5.2.4. Communication Agent:

The AI Assistant in SmartBreastCare highlights common breast cancer symptoms like lumps and nipple discharge, promoting early detection and advising consultation with healthcare professionals. It uses Reinforcement Learning and Ensemble Methods to enhance decision-making and patient interaction accuracy.

5 Conclusion.

The Agentic AI-based breast cancer detection system provides high diagnostic accuracy and reliable recommendations, significantly improving clinical workflows. The frontend interface ensures easy accessibility for both patients and clinicians, making the process smooth and user-friendly. The AI-powered chat assistant further enhances patient engagement and provides real-time insights into the diagnostic process. Further refinements, including expanding the dataset and enhancing explainability, are planned to improve the model's robustness and integration into clinical settings.

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